

HR Analytics: Employee Attrition Report

Comprehensive Workforce Stability & Performance Analysis

1. Project Overview

Project Title: IBM HR Analytics

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Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, Excel

Dataset: IBM HR Analytics Dataset (1,470 Records, 35 Attributes)

Employee attrition represents a critical challenge for organizational productivity and stability [cite: 1, 2]. This project identifies the key factors influencing turnover to help design effective retention strategies by analyzing employee demographics, job roles, and performance [cite: 2].

2. Executive Summary

The analysis reveals a current attrition rate of **16.12%** [cite: 1]. While the majority of employees remain with the company (83.88%), the loss of approximately one-sixth of the workforce incurs significant recruitment and training costs [cite: 1, 2]. Key drivers identified include age demographics, job satisfaction levels, and financial compensation structures [cite: 1, 2].

3. Data Methodology & Cleaning

The analysis followed a structured data science pipeline as outlined in the project documentation [cite: 1, 2]:

- **Data Collection:** Utilization of the fictional IBM HR dataset containing 1,470 records and 35 attributes [cite: 1, 2].
- **Data Integrity:** Verified 0 missing values and 0 duplicates, ensuring accuracy and consistency [cite: 1, 2].
- **Analysis:** Conducted Exploratory Data Analysis (EDA) and built predictive models (Decision Trees) to identify patterns [cite: 2, 3].

4. Key Findings

4.1 Demographic Trends

- **Age:** Peak attrition density occurs among employees in the 25–35 age group [cite: 1].
- **Gender:** The majority of employees are male, and they exhibit a higher overall count of attrition [cite: 1].
- **Department:** Most employees are concentrated in the Research & Development department [cite: 1].

4.2 Retention Indicators

Metric	Average/ Percentage	Insight
Attrition Rate	16.12%	Represents the proportion of employees who left during the period [cite: 1].
Average Tenure	7.01 Years	Indicates long-term employee comfort prior to turnover [cite: 1].
Work-Life Balance	Ordinal (1-4)	Balance levels significantly impact retention decisions [cite: 1].

5. Predictive Modeling Insights

Based on the Decision Tree analysis in the project notebook [cite: 3]:

"Key factors influencing attrition include lower monthly income, low job involvement, and the absence of overtime opportunities or excessive workload imbalances."

6. Conclusion & Recommendations

To mitigate turnover, the following actions are recommended based on the generated insights [cite: 1, 2]:

1. **Targeted Retention:** Focus on the 25-35 age demographic with tailored career development [cite: 1, 2].
2. **Satisfaction Monitoring:** Regularly assess job involvement and environment satisfaction to prevent early exits [cite: 1, 2].
3. **Compensation Benchmarking:** Review monthly income levels, particularly for roles with high turnover risk [cite: 2].

Sources: [1] IBM HR Analytics Problem Statement; [2] Internship Work Report (Vedant Parmar); [3] Employee Attrition Analysis Notebook.